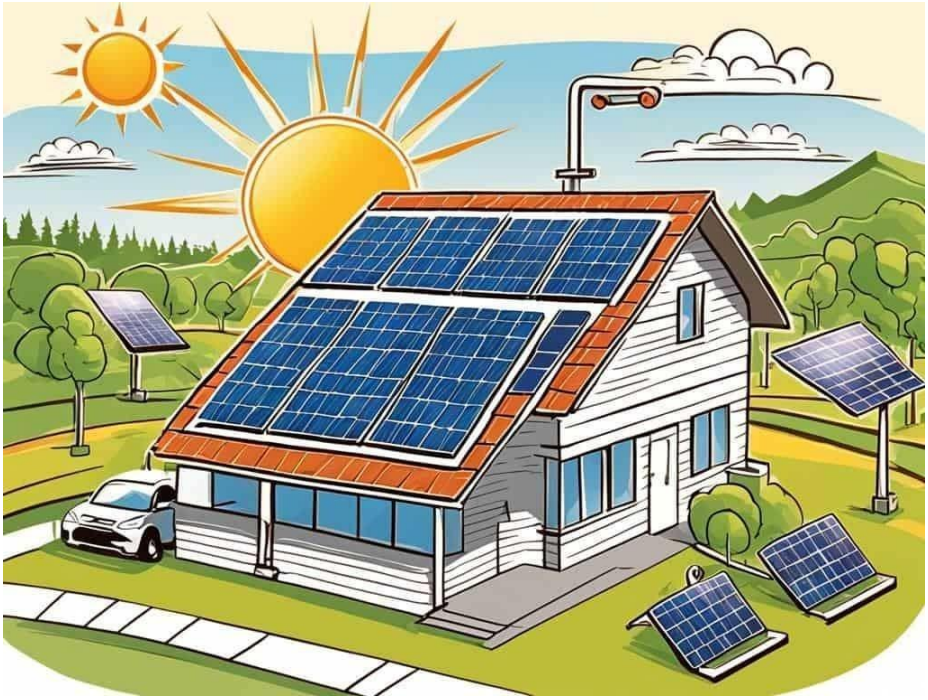
PH-101 PROJECT REPORT



SOLAR TRACKING SYSTEM USING ARDUINO

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AIM OF THE PROJECT

Designing and constructing single axis solar tracking system using Arduino and LDR.

MATERIALS USED

- **Arduino UNO board: It is a microcontroller that takes input from the Light dependent resistors (LDRs) and processes them according to the provided code. It then gives output in the form of instructions to the Servo motor to rotate.**

- **Solar panel: Demonstrative solar panel**

- **SG90 Servo motor: It is used to rotate and re-align the solar panel as required.**

- **LDR sensors: 2 LDRs are placed on either side of the solar panel, acting as sensors**

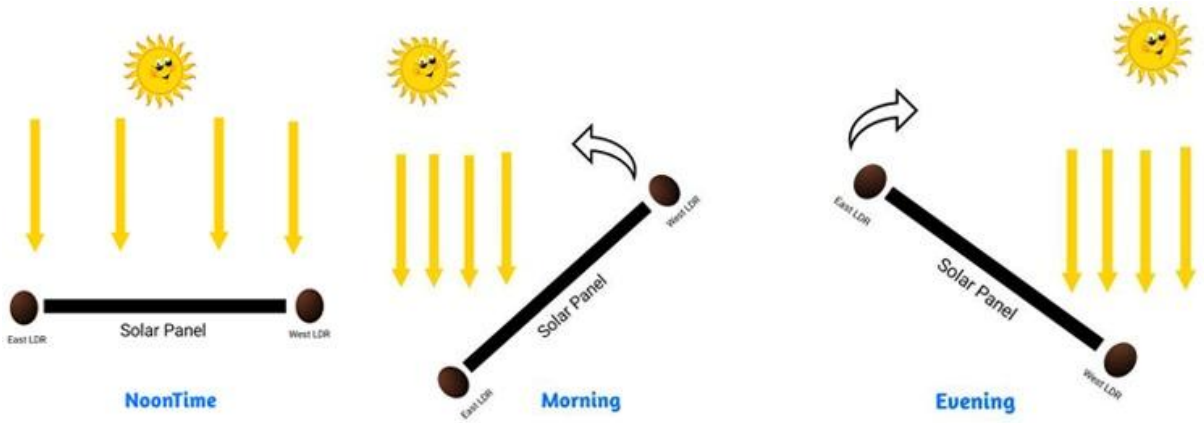
- **10k resistors: To scale the values fed through the LDR correctly.**

- **Jumper wires: To connect every part of the circuit.**

- **Wood Piece: To provide support to the Servo motor and act as a base.**

WORKING PRINCIPLE OF LDR BASED SOLAR TRACKING

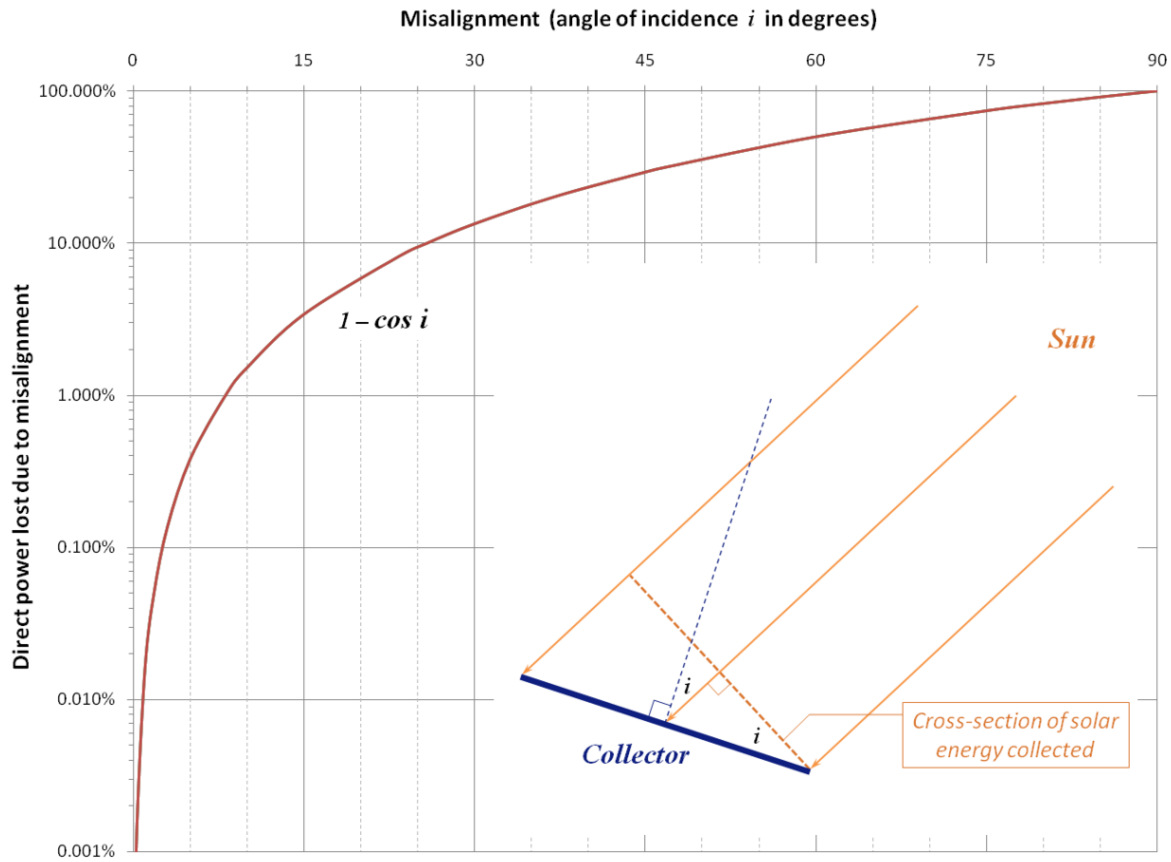
The light-sensing system operates by using two or more light-dependent resistors (LDRs) positioned on either side of the solar panel. These LDRs exhibit high resistance in low light conditions and low resistance in bright light. When sunlight hits the LDRs, the one receiving more light has a lower resistance, creating a difference in current across them. This difference is detected by an Arduino microcontroller, which reads the signals from the LDRs and calculates the variations in light intensity. Based on these calculations, the Arduino determines the direction in which the solar panel should move to maximize exposure to the light source. It then sends a command to a servo or stepper motor attached to the panel, rotating it towards the LDR with the highest light intensity. This setup enables the solar panel to continuously track the sun's movement throughout the day, optimizing energy capture by ensuring the panel always faces the sun.



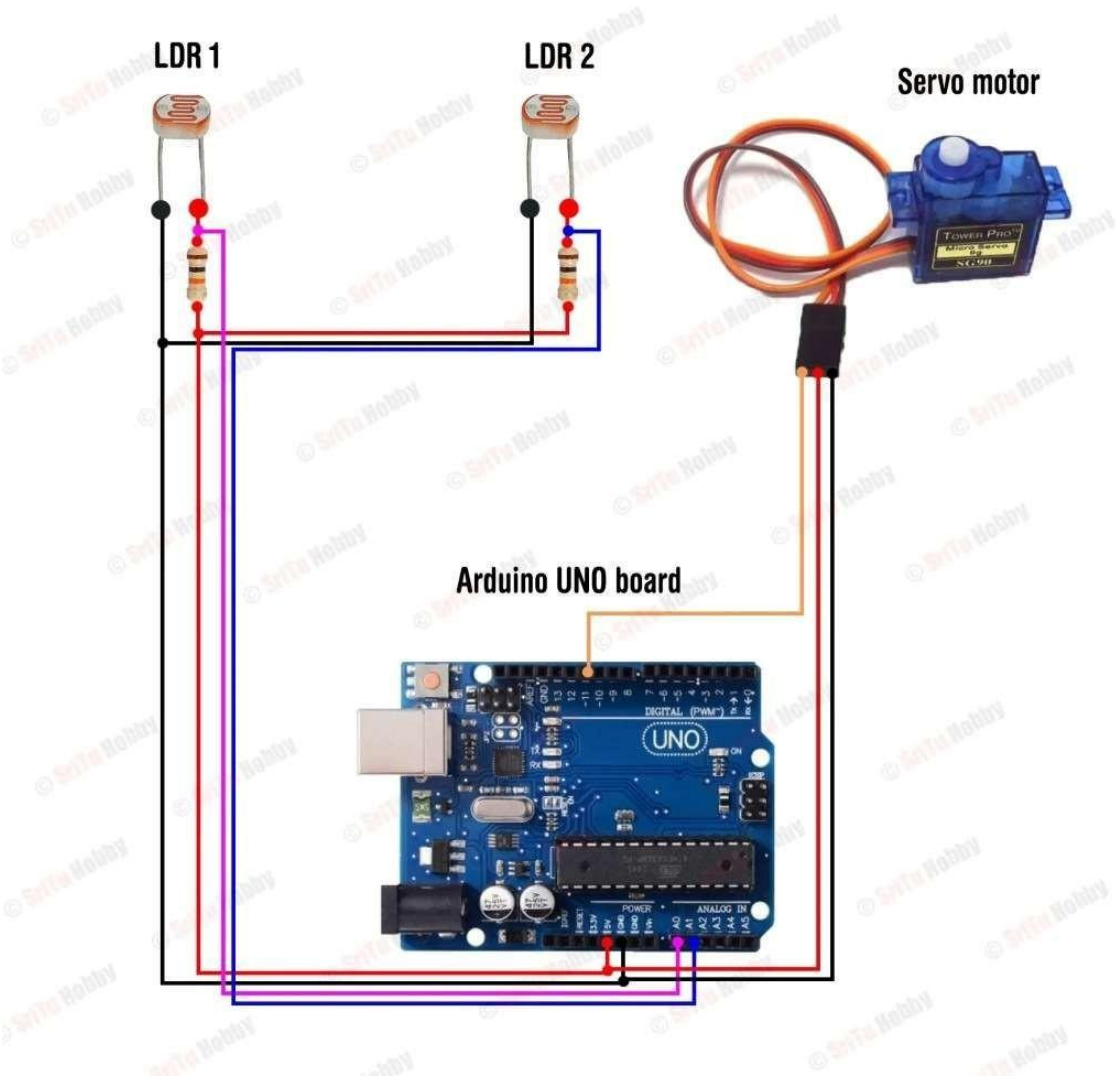
IMPORTANCE OF SOLAR TRACKER

Solar Trackers are used to re-orient solar panels in the direction where a high amount of solar intensity can be captured. Throughout the day the sun moves from east to west, and a fixed solar panel will only be able to generate 60-70% of the power when compared to a movable solar panel.

Single axis trackers move the solar panel about 1 axis, using sensors to orient the solar panel in the direction perpendicular to the sun's rays. This type is the most widely used solar tracking system. Dual axis solar trackers also exist, however they improve the efficiency of solar panels by very minute amounts. They do not justify the cost required to implement them.



CIRCUIT DIAGRAM



CHALLENGE FACED

LDR Sensitivity: LDRs are sensitive but not very precise, which can result in inaccurate readings, especially in low light or mixed-light conditions. This might lead to incorrect tracking or slower adjustments. We faced issues in checking the values of both the LDRs because of their sensitivity. Even putting our hand in front of the LDR leads to rotation of the motor. So we had to do it very carefully so that accurate tracking takes place.

1. **Books and Journals:**

REFERENCES

Mazidi, M. A., Naimi, S., & Naimi, S. (2013). The Arduino Microcontroller: A Hands-on Approach. Pearson. - A comprehensive guide on Arduino programming and electronics.

Goswami, D. Y., Kreith, F., & Kreider, J. F. (2000). Principles of Solar Engineering. CRC Press. - Detailed discussion on solar energy systems and engineering principles.

2. **Online Tutorials:**

Arduino Guide. (n.d.). LDR Sensor Interfacing with Arduino. Arduino Project Hub. Retrieved from Arduino.cc - An online resource for Arduino projects and sensor interfacing.

Intractable. (n.d.). How to Build a Solar Tracker. Retrieved from Instructables.com - Step-by-step guides for DIY electronics projects.

CONCLUSION

This solar tracking experiment shows how an LDR-based system improves solar panel efficiency by continuously aligning it with the sun, capturing more energy than a stationary panel. The setup, using Arduino and basic sensors, is cost-effective and adaptable. Although it has limitations, like sensitivity to low light, future upgrades such as dual-axis tracking and better sensors could further

enhance its reliability and efficiency for sustainable energy applications.